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A geomagnetic secular variation study (31 000–19 500 bp) in Western Canada

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Summary. Palaeomagnetic results from 212 horizons spread evenly through an 18 m sedimentary sequence in southern British Columbia are reported. Radiocarbon ages suggest that the sequence spans the interval from 31 200 to 19 500 yr bp. No evidence for any large geomagnetic excursions (such as the so-called Mono Lake Excursion) is found, but a distinctive pattern of 'normal' secular variation is observed with declination and inclination swings of 45° and 25° peak to peak amplitude respectively. For the most part the secular variation consists of low amplitude oscillations about the field vector of a geocentric axial dipole expected at the site latitude, but three relatively large perturbations occur at approximately 4000 yr intervals. These perturbations systematically bias the overall mean to shallow inclinations and easterly declinations in a manner reminiscent of the spatially non-isotropic secular variation model proposed by Cox. The bias involved is about 6° in declination and 3° in inclination (overall mean $D = 5.8°$E, $I = 64.2°$, $\alpha_{95} = 0.9°$, $N = 212$ horizons), which leads to a pole which is both 'far-sided' and 'right-handed'. If the horizons involved in the three major perturbations are eliminated the mean direction ($D = 1.2°$E, $I = 67.2°$, $\alpha_{95} = 0.8°$, $N = 125$ horizons) does not differ from that of a geocentric axial dipole despite the small cone of confidence.

1 Introduction

During the last decade there has been a renewed interest in the long-term secular variations of the geomagnetic field. Magnetic observatories provide one important source of data for such investigations, but these are severely restricted by the fact that they cover relatively short time intervals. For earlier times it is necessary to use archaeomagnetic and palaeomagnetic data. Many of the problems involved require long time sequences of data, and to this end palaeomagnetists have increasingly turned to investigations of suitably deposited Quaternary sediments. This paper reports palaeomagnetic results from such a sequence spanning about 12 000 yr, between 31 000 and 19 000 yr bp.

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2 Geological and sampling details

During the late Quaternary the mid-latitudinal zones of the Earth experienced a series of glacial and interglacial episodes. In southern British Columbia and north-western Washington State these are usually referred to as the post-glacial (0–10 000 bp), the Fraser Glaciation (20 000–10 000 bp), the Olympia Interglacial (43 000–20 000 bp) and the Okanagan Glaciation (?–43 000 bp) (Westgate & Fulton 1975). The site of the present investigation (50° 7′ 55″N, 118° 51′ 45″W) is a stream cut escarpment on Bessette Creek, 36 km east of Vernon, south-central British Columbia, known locally as Riggins Road, where floodplain sediments deposited during the Olympia Interglacial are now exposed.

The sequence is described by Westgate & Fulton (1975), and illustrated in Fig. 1. There are approximately 20 m of mainly fine grained organic muds and silts above the present stream level. The top of a basal gravel layer is used as the origin of our elevation scale. Two volcanic ash bands, identified by Westgate & Fulton as the Riggins Road and Cherryville

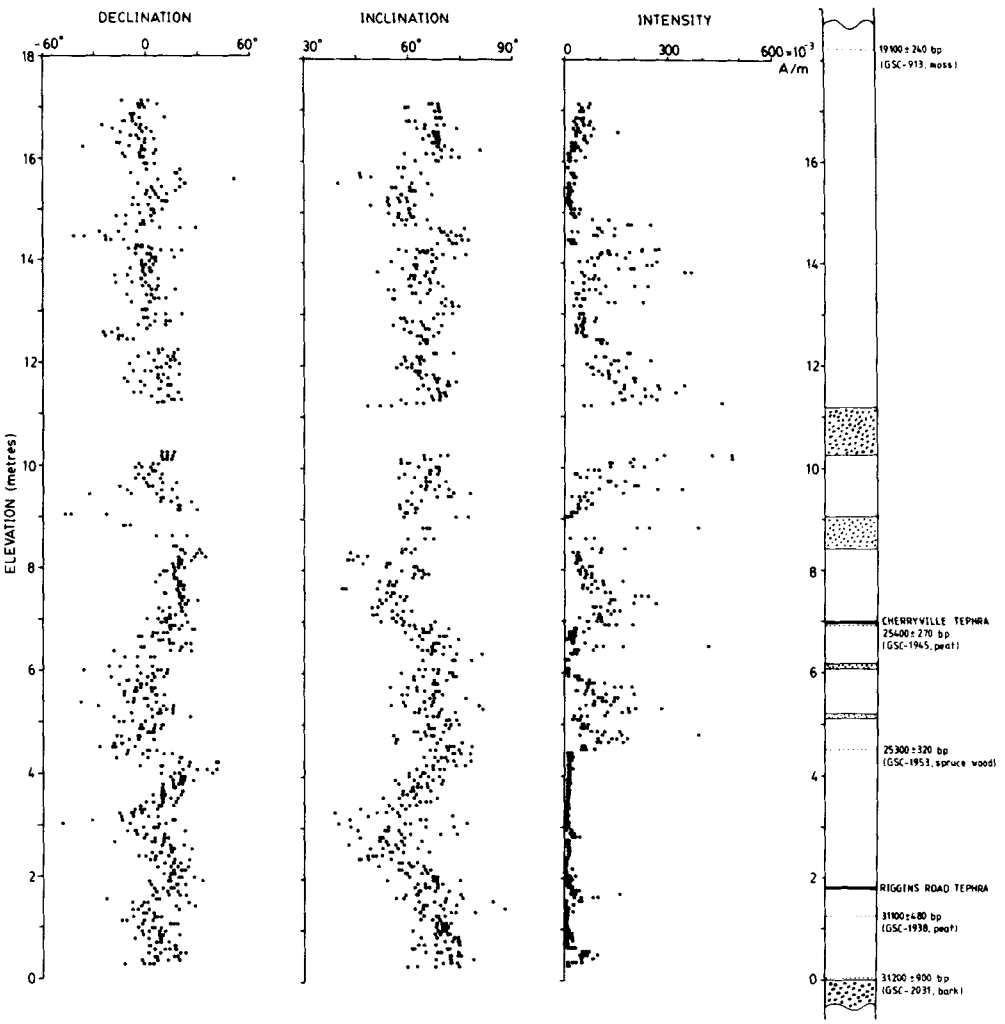


Figure 1. Plots of declination, inclination and intensity of natural remanent magnetization (NRM) of all samples, and simplified lithology, plotted against original elevation in the exposure. Stippled sections represent sand and gravel layers.

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Tephra, and dated by them at approximately 30 000 and 25 000 yr bp, occur at 1.80 and 7.00 m respectively. Our sampling is interrupted by two gravel bands, one dark grey and very wet between 8.42 and 9.04 m, the other rusty in colour and drier between 10.25 and 11.20 m. The whole sequence is overlain by coarse sand and boulders representing the subsequent Fraser Glaciation.

The lowermost 7 m of the section have been sampled previously for palaeomagnetic work by Oberg & Evans (1977), who collected two samples per horizon at approximately 19 cm stratigraphic intervals. We resampled this part, and a further 11 m using a finer interval of 7 cm between horizons and taking three or four samples per horizon.

Most specimens were collected by pushing or tapping small (~ 2 cm) plastic boxes into a prepared near vertical surface, but at nine horizons, in very compact silt layers, it was necessary to carve protruding stumps from the face, over which the cubes could be fitted. All the samples were fully oriented using a Brunton compass-clinometer, before removing, sealing and wrapping in moist paper towels to prevent drying and deformation.

Forty magnetic Sun bearings made throughout the sampling period yielded an average local declination of 21.7 ± 0.8° (sd) E. The small variance and agreement with the regional average of 22.3°E (1975.0 Canadian Isogonic Chart), indicates the precision and accuracy of the method.

3 Magnetic measurements

3.1 NATURAL REMANENCE (NRM) AND ALTERNATING FIELD DEMAGNETIZATION

Prior to, and between, measurements the samples were stored in a magnetically shielded room, in which the field is reduced to less than 50 gamma, to minimize any viscous magnetizations. All measurements were made on a Digico complete result balanced fluxgate magnetometer (Molyneux 1971), with an overall noise level of less than 0.1 mA m^{-1} per sample (or 5×10^{-10} A m^2 total magnetic moment). Alternating field (af) demagnetization was carried out using the two-axis tumbling system described by Murthy (1969).

Fig. 1 shows the NRM data of all the samples. Although the intensity varies by two or more orders of magnitude, with notable changes occurring at around 4.50 and 15.0 m, this does not appear to affect the quality of the directional record, or the spread of directions within a horizon. Of the 212 horizons sampled, Fisher's precision parameter, k, is less than 100 in only 20 cases, and is greater than 1000 in 26 cases.

As a further test of the stability and reliability of the remanence 19 pilot samples, taken either singly or in pairs from throughout the section, were subjected to routine stepwise af demagnetization, in peak fields of 2.5, 5, 10, 15, 20, 30, 40 and 50 mT. Fig. 2 shows vector component diagrams for four typical samples. All 19 samples behaved very similarly and had median destructive fields ranging from 10 to 33 mT, with a mean of 22 mT. In some cases the initial steps removed a small component and resulted in small, but measurable, angular shifts, generally away from the present field direction at the sampling site. Between 10 and 30 mT directions stabilized (mean angular shift = 3°, maximum = 9°). Some samples remained stable beyond 30 mT but others began to diverge widely. The magnetic vectors removed between 0 and 5 mT ($N = 19$, $D = 349.0°$E, $I = 74.2°$, $k = 5.1$, $\alpha_{95} = 16.6°$) are rather scattered but systematically steeper than the 'cleaned' results, as would be expected if a recent viscous component is present, the local inclination being currently 5° steeper than that of an axial dipole field. However, the vectors removed between 5 and 10 mT are much more tightly grouped ($N = 19$, $D = 5.4°$E, $I = 62.8°$, $k = 38.3$, $\alpha_{95} = 5.5°$) and do not differ from the overall mean of the entire section obtained after cleaning. We conclude that a

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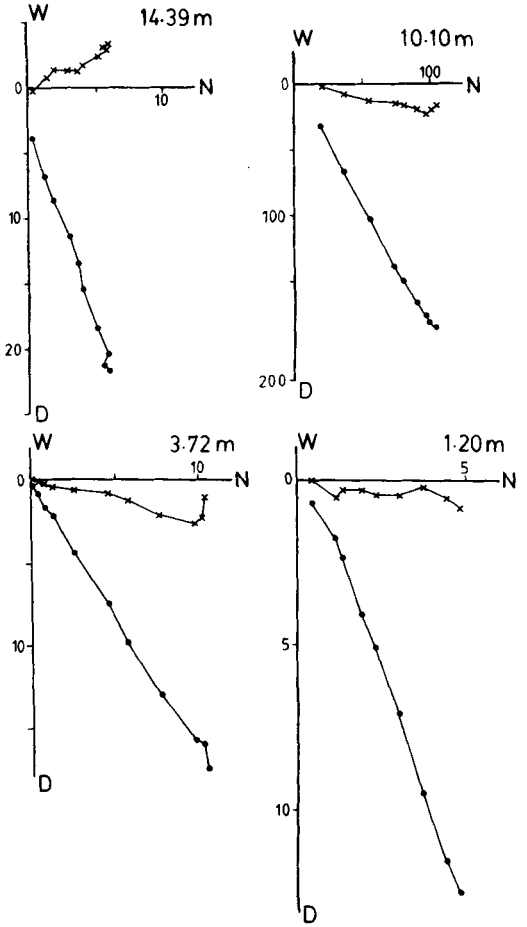


Figure 2. Vector component diagrams showing the behaviour of four typical pilot samples on af demagnetization. The axis units are mA m^{-1}. The samples are identified by their elevation in the exposure. The horizontal component of remanence is plotted as crosses, the vertical component in the north–south plane as dots.

primary remanence has been isolated following removal of a small viscous secondary component acquired in recent times.

On the basis of the pilot studies all remaining specimens were subjected to af demagnetization at 10 mT, and the results obtained form the basic dataset to be analysed below. The cleaning did not significantly improve the within horizon grouping, but did result in an overall inclination shallowing of about 1°, to give a mean direction over the whole section of $N = 212$, $D = 5.8°\text{E}$, $I = 64.2°$, $\alpha_{95} = 0.9°$.

3.2 THERMAL DEMAGNETIZATION AND ISOTHERMAL REMANENCE

The intensity of remanence was measured as a function of temperature for four specimens, using a Digico high temperature spinner magnetometer. The blocking temperature spectra are all virtually identical, and imply a Curie temperature of approximately 450°C, corresponding to a titanomagnetite of composition 80 per cent magnetite, 20 per cent ulvospinel.

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A further four specimens were given isothermal remanent magnetizations (IRMs) in successively higher fields up to approximately 1 T. Again the results were virtually identical, rising sharply to 0.1 T, and saturating in a field of approximately 0.3 T. Taken together the thermal and IRM results suggest that the remanence in these samples is carried by titanomagnetite grains, haematite being either absent or negligible. This in turn suggests that the remanence is depositional or post-depositional in nature, and free of later chemical overprints.

4 Chronology

A crucial element of secular variation studies, both from the geomagnetic aspect, and for the purpose of correlating records from different localities, is the establishment of a reliable chronological scale. In the present case the relevant information available is limited, and its interpretation not totally unambiguous, but, bearing this in mind, a useful, approximate time-scale can be established.

Firstly the two gravel bands between 8.42 and 9.04 m and 10.25 and 11.20 m must be dealt with. The upper band is very distinctly different from the silt above and below, and is visible, at approximately the same thickness along the 100 m or so exposed at the site. Declination, inclination and intensity of remanence are continuous across the band, however (Fig. 1), suggesting that it represents a very short episode of time, during which the magnetic field changed little. Similarly, although the few samples it was possible to collect within the lower gravel band yielded scattered directions and intensities, the results from the silt above and below match well. Thus it was decided to discard both gravels as representing episodes of short duration, and to plot the remainder of the record continuously.

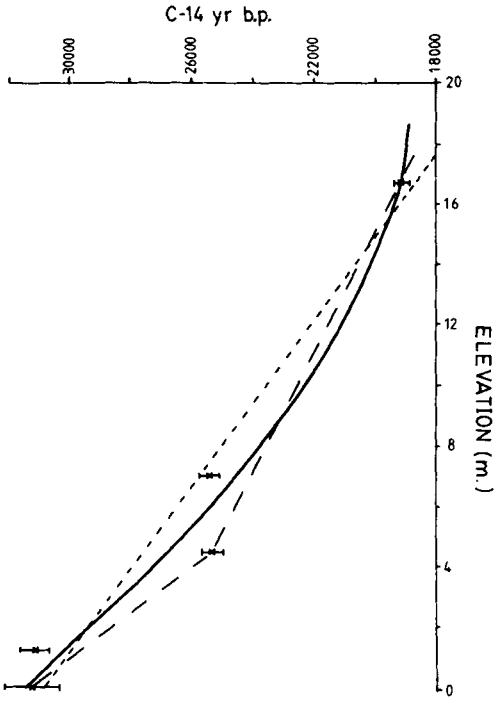


Figure 3. Radiocarbon age estimates available on the section and theoretical deposition rates calculated from them, against elevation scale adjusted for the removal of the gravel bands. Long dashed line = linear interpolation between three selected dates (see text), short dashed line = linear regression through all five dates, solid line = quadratic regression through all five dates.

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The positions of the five C-14 age determinations available are shown in Fig. 3, the vertical scale having been adjusted by removal of the gravels. Since these determinations were used by Westgate & Fulton (1975) in dating the tephra, we cannot treat the tephra ages as independent controls. Clearly the lowermost two dates are in disagreement, as are the estimates at 4.50 and 7.00 m. This leads to two approaches to the development of a time-scale:

(a) Only those dates considered more likely to be correct are used, the others being discarded. The C-14 clock begins when biological material dies and ceases to absorb more radiogenic carbon from the atmosphere. Hence, organisms which die long before they are buried in the sediment will yield anomalously old age determinations. Age determinations erring on the young side are less common, but have on occasion been explained by the percolation of modern groundwater, contaminating a sample with recent carbon (Olsson & Eriksson 1965). On this basis it might be logical to disregard the two dates made on peat fragments and accept those on bark at 0.0 m, wood at 4.50 m, and moss at 18.20 m, as these are younger and less likely to have had a long pre-depositional history. This was the approach taken by Oberg & Evans (1977) who interpolated linearly between the two lower dates, and extrapolated up to the 7 m level. Similar linear interpolation in two stages between the three dates we have here would result in an unrealistic break in sedimentation rate by a factor of three at 4.50 m (Fig. 3). Clearly a smooth curve fitted to the data would be more realistic, but is unjustified with only three points.

(b) Our preferred approach uses all five estimates on an equal weight basis. Fig. 3 also shows the best fitting straight line and quadratic through the five points. Linear regression makes the assumption of a uniform deposition rate over the whole 12 000 yr period – an assumption which is not supported by the data themselves. Quadratic regression tends to force the curve through the uppermost point, then through the mid-points of the other two pairs. The uppermost date is probably quite reliable, as it is just below the till, and the onset of the Fraser Glaciation has been placed at 19 000–20 000 bp at other sites in the area also. We consider this the best time-scale possible with the currently available data. It yields ages of 31 200 and 19 500 bp for the top and bottom of the palaeomagnetic section and ages of 29 500 and 24 600 bp for the Riggins Road and Cherryville Tephra respectively, in excellent agreement with the estimates of Westgate & Fulton (1975). It implies a sedimentation rate of 0.085 cm yr^{-1} at the base and 0.28 cm yr^{-1} at the top of the section, corresponding to 82 and 25 yr respectively for the 7 cm sampling intervals.

A final factor which should be appreciated with regard to radiocarbon dates, but which cannot be quantified, is the probable difference between the C-14 time-scale (yr bp) and the calendar (yr BP). This difference arises due to the time varying radiogenic carbon content of the atmosphere with changing cosmic ray flux and magnetic field strength. Dendrochronological calibration curves cover the last 7000 yr (Clark 1975), and in some cases discrepancies of as much as 1000 yr exist between the two scales. It would be unreasonable to assume that similar differences may not have occurred in earlier times also.

5 Discussion

Fig. 4 shows the horizon mean directions, after af cleaning at 10 mT and removal of the two gravel bands, plotted against elevation. In Fig. 5 the same data are presented, after passing a three point vector running average over the horizon means, and plotted against age using the quadratic regression time-scale described above. Also shown are the 95 per cent confidence limits, calculated in the same three horizon window about each point. The use of a three horizon window in calculating alpha-95's is a trade-off which permits the

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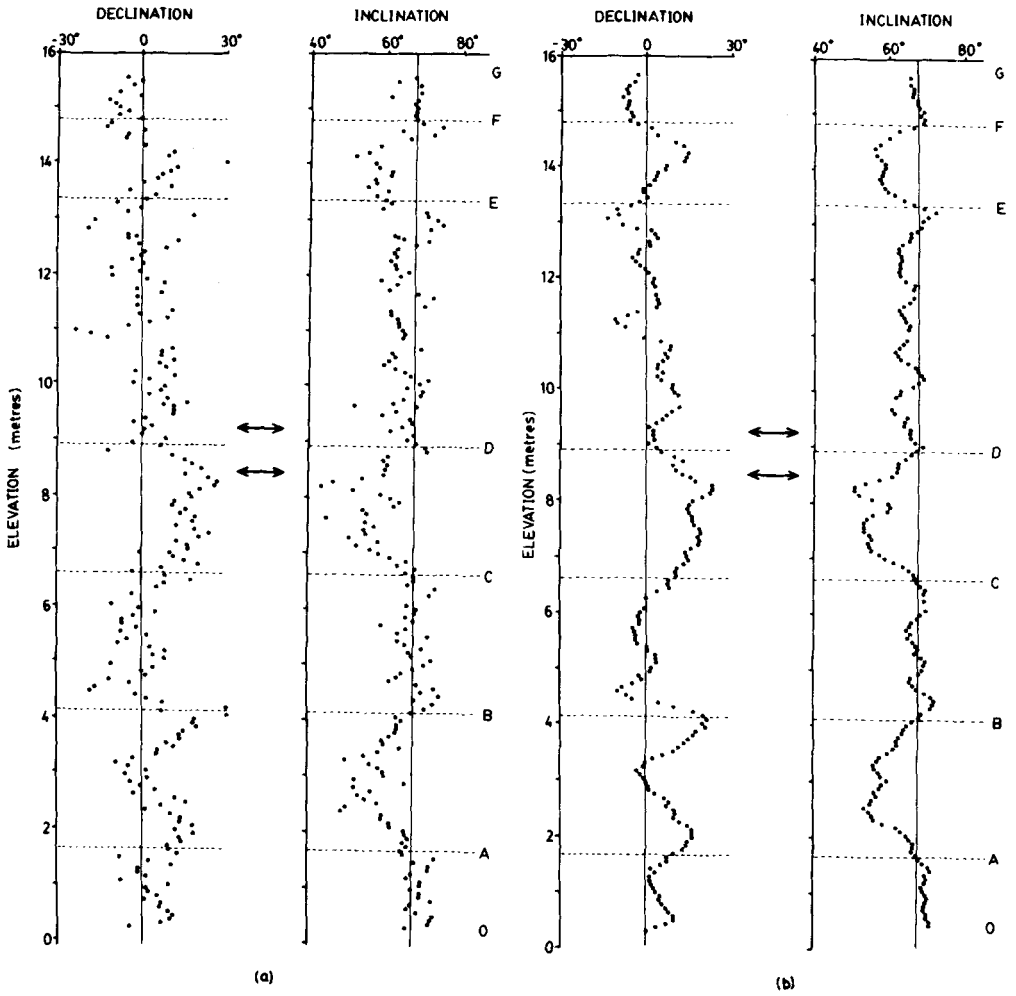


Figure 4. (a) Horizon mean declination and inclination. The arrows show the positions at which the elevations have been adjusted by removal of the two gravel bands. (b) As (a) after passing a three point vectorial running mean over the dataset.

consideration of a statistically significant number of data points at each calculation without including too large a contribution from between horizon differences in direction, i.e. the actual secular variation.

Discussion of the results summarized in Fig. 5 can be conveniently broken down as follows.

5.1 GEOMAGNETIC EXCURSIONS

The whole record consists of 'normal' secular variation swings of peak to peak amplitude 45° and 25° in declination and inclination respectively. The amplitude and character of these swings are very similar to those recorded in lake sediments from Britain (Turner & Thompson 1981), Europe (e.g. Creer 1981) and North America (Lund & Banerjee 1979), at similar latitudes spanning the last 10 000 yr.

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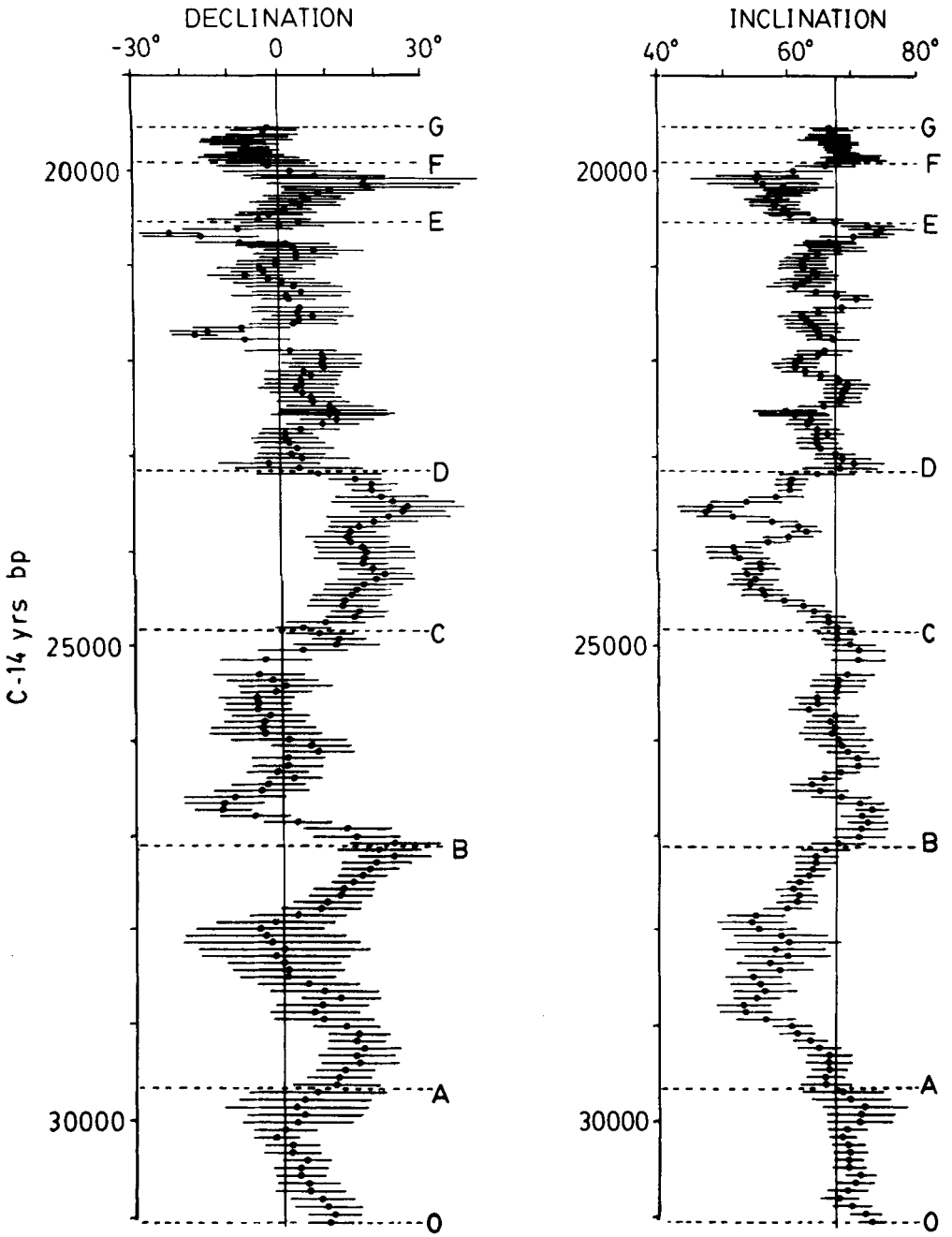


Figure 5. Horizon mean declination and inclination plotted against the quadratic regression time-scale (Fig. 3). Ninety-five per cent confidence limits and running means are calculated using a three-point moving window, A–B, C–D and E–F delineate the three major swings (see text).

There is no evidence of a larger scale 'excursion' of the type observed at Mono Lake, California, only 12° south of our site (Liddicoat & Coe 1979). At Mono Lake the inclination excursion, accompanied by extreme westerly declinations, was originally dated at 24 000–25 000 bp. However, there is a growing bank of data from western North America between

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c. 35 000 and 20 000 bp recording purely 'normal' secular variations: no excursion has been found at Clear Lake, California between 29 000 and 21 000 bp, (Verosub 1979), Pyramid Lake, Nevada between 36 000 and 25 000 bp (Verosub, Davies & Valastro 1980) or here at Bessette Creek between 31 000 and 19 500 bp; Liddicoat & Coe considered the excursion a feature of the non-dipole field rather than a world-wide reversal, and modelled it by an eccentric dipole at 0.28 Re, drifting from north to south of the site and reversing in sign through the excursion. Any such feature should certainly have been recorded at these other sites. If the feature recorded at Mono Lake reflects a true geomagnetic phenomenon, one is forced to conclude that the dating control is in appreciably more error than previously thought, and that the excursion occurred before 36 000 bp, as has also been suggested by Verosub *et al.* (1980). It is interesting to note that proponents of the Laschamp event have also pushed its age back into this period (Hall & York 1978; Gillot *et al.* 1979) from the 8000–20 000 bp originally suggested (Bonhommet & Zahringer 1969). Even if not a world-wide reversal, one could speculate that the same drifting non-dipole source might have produced the two excursions within a relatively short time of one another.

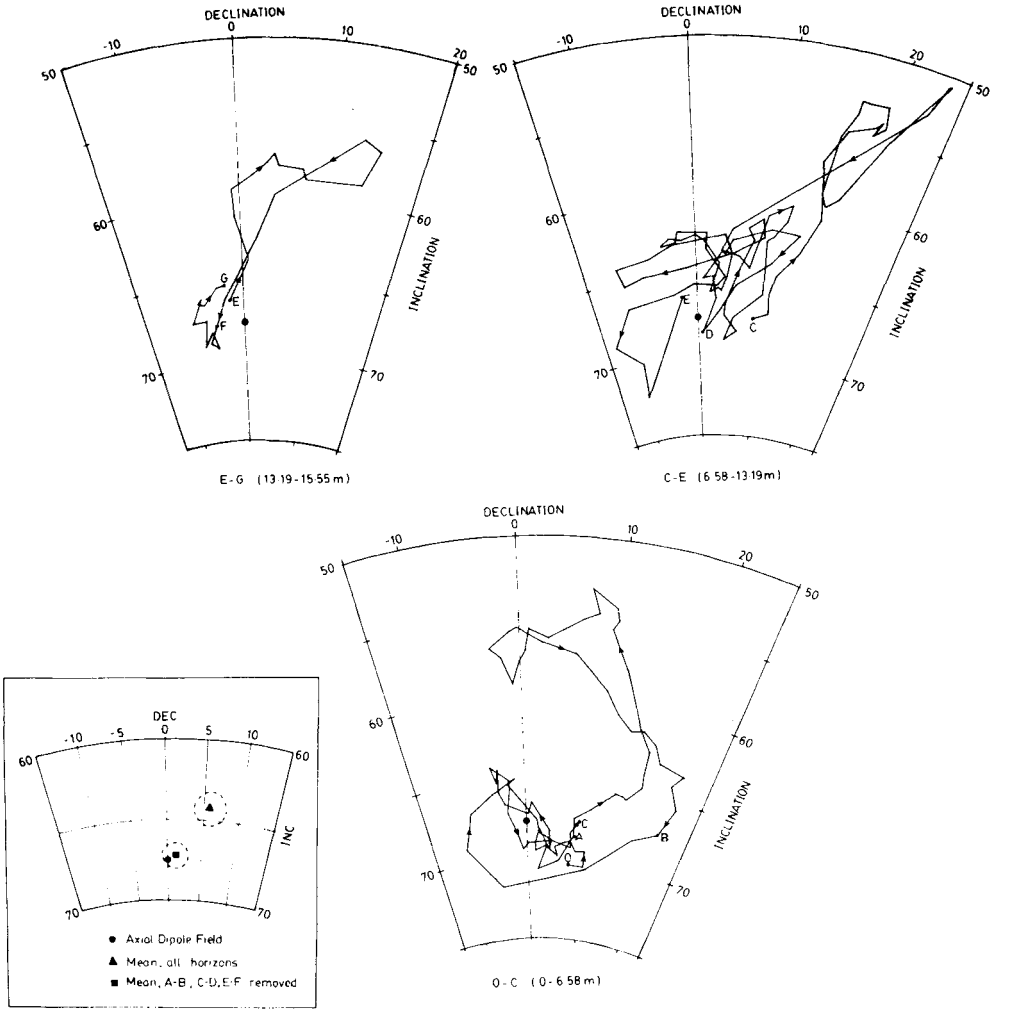


Figure 6. Bauer plots (equal area projections) of the magnetograms of Fig. 5. The lettering of the sections refers to Fig. 5. Large dot = geocentric axial dipole field direction. The insert shows how the removal of the major swings moves the mean direction of the section to the ADF.

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5.2 'QUIET' AND 'NOISY' SECULAR VARIATION

Although there are no indications of any large 'excursions' in this record, inspection of the inclination magnetogram (Fig. 5) reveals that the secular variation signature consists of intervals of relatively low amplitude, high-frequency variations (O–A, B–C, D–E, F–G) interrupted by three major swings to shallow inclinations (A–B, C–D, E–F). During these swings the field vector tends to be biased to the east, a feature which is better illustrated by the Bauer plots given in Fig. 6.

This pattern of low amplitude variations interspersed with relatively large perturbations parallels the 'quiet' and 'noisy' secular variation intervals found in detailed studies of Hawaiian lavas summarized by Cox (1975). Lack of an adequate time-scale for the Hawaiian data limits what can be deduced about the temporal behaviour of these 'noisy' intervals, but the Bessette Creek data are suggestive of a recurring phenomenon, superimposed on a principally axial dipole main field. We return to this point below, after discussion of another important feature of the Bessette Creek data, namely the bias towards shallow inclinations and easterly declinations.

5.3 BIASED SECULAR VARIATION

It is clear from Fig. 5 that the difference between the mean direction ($D = 5.8°$E, $I = 64.2°$) and the axial dipole field (ADF, $D = 0°$, $I = 67.5°$) results from the three low inclination swings (A–B, C–D, E–F). If these intervals are excluded one obtains a mean direction of $D = 1.2°$E, $I = 67.2°$ ($\alpha_{95} = 0.8°$, $N = 125$ horizons), which does not differ significantly from the ADF despite the small cone of confidence (Fig. 6).

The records from Pyramid Lake, Nevada (Verosub *et al.* 1980) and Clear Lake, California (Verosub 1979), mentioned above are not detailed enough to permit swing to swing correlations, but it is interesting at this stage to compare their mean directions. At Pyramid Lake the mean, between 36 000 and 25 000 bp is $D = 5.2°$E and $I = 50.1°$, $9.1°$ shallower than the site ADF and eastward biased, as at Bessette Creek. The Clear Lake record consists of a number of unoriented sections, but the mean inclination is again appreciably shallower than the ADF. Although such shallow inclinations may be attributed to inclination error (King & Rees 1966), this coincidence of shallow inclinations and easterly declinations at several sites strengthens the case for a real bias of the main dipole field.

The effect of the major swings in our data is then to displace the mean almost 6° east of and 3° shallower than the ADF. This displacement leads to a VGP at 84.3°N, 16.0°E, which is 'far-sided' and 'right-handed' in the sense introduced and developed by Wilson (1971). It is likely that the 11 700 yr spanned by our dataset are insufficient to cover the full spectrum of secular variations, and that this bias may not represent a persistent long-term phenomenon. However, Wilson maintains that the far-sided effect has existed since upper Tertiary times. He explained it in terms of a small northward shift of the central axial dipole, which is formally equivalent to the existence of a permanent quadrupole moment (Merrill & McElhinny 1977). The northward shift implied by global palaeomagnetic data is of the order of a few hundred kilometres, corresponding to a ratio of $g_1^0/g_2^0 \simeq 20$ for the axial dipole and quadrupole coefficients.

As discussed above, our observed pattern of secular variation is reminiscent of Cox's (1975) discussion of Hawaiian lava flows in that the main perturbations from the ADF are systematically biased towards shallow inclinations. However, there is an important difference between the present data set and the model proposed by Cox. He concludes that global data imply a bias towards shallow inclinations which is geographically restricted, being most pronounced at the equator and decreasing to zero at latitudes between 40° and 60°, beyond

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which steep inclinations prevail. On this basis one would expect no significant inclination bias in our data, whereas we observe an inclination shallowing of $3.3 \pm 0.8°$. A more recent global summary by Merrill & McElhinny (1977), using 221 data points as opposed to Cox's 135 suggests that shallow inclinations are expected at all latitudes, decreasing monotonically from equator to pole. Thus it appears that Cox's suggestion of non-isotropic secular variation provides an appealing physical mechanism but that the bias may have the same sign at all latitudes.

The eastward bias, or right-handed effect is much more troublesome – if it really exists it implies a long-term toroidal field at the Earth's surface. Merrill & McElhinny claim that Wilson's interpretation of a world-wide eastward bias results from inadequate global coverage of data. They conclude that there is a weak sinusoidal variation of declination along lines of latitude, producing an eastward bias (right-handed pole) between 60°W and 120°E and a westward bias between 120° and 300°E. The eastward bias we observed at a longitude of 241.1°E thus lends no support for the model of Merrill & McElhinny.

5.4 SPECTRAL ANALYSIS

Spectral analysis has been performed on the vector dataset of declination and inclination, interpolated to 50 yr time intervals and mapped on to the complex plane (Denham 1975). A Fast Fourier Transform technique was employed, after applying a cosine taper and extending the dataset to 1000 points, by adding zeros (see, e.g. Kanasewich 1981, chapters 3 and 9). This smooths the long-period end of the spectrum, but tests show that it does not significantly affect the resolution or relative amplitudes of the peaks.

The power is roughly equally divided between positive and negative frequencies (Fig. 7), reflecting no overall preference towards either clockwise or anticlockwise looping of the magnetic field vector. The dominant peaks at 4000 yr are associated with the repetition times between the three major swings (Fig. 5). Closer examination of Fig. 5 shows that the 2400 and 1850 yr peaks are caused by the uppermost major swing E–F and variations within and between the two lower swings. The other, higher-frequency peaks originate from the low amplitude, between swing variations, e.g. D–E, and will not be discussed further.

Oberg & Evans (1977) carried out detailed spectral analysis on their much smaller dataset, corresponding to O–C (Fig. 5). They reported anticlockwise 5000 yr amd clockwise 1900 yr

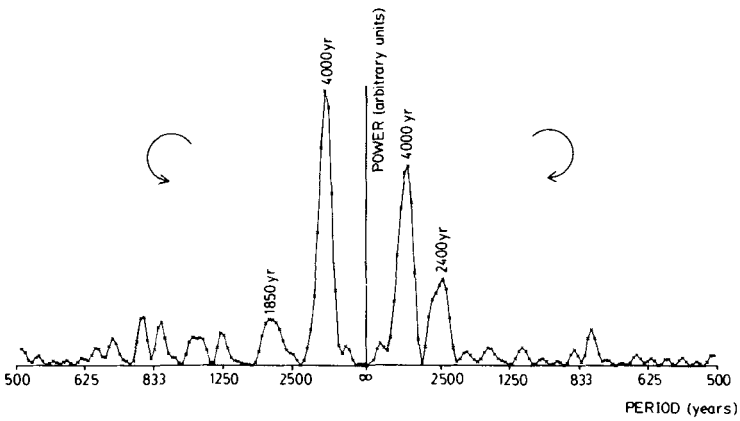


Figure 7. Power spectrum of the entire declination and inclination sequence combined as a complex number series and extended to 1000 points. Peaks on the right- and left-hand sides of the diagram correspond to predominantly clockwise and anticlockwise looping of the magnetic field vector respectively.

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periodicities which they associated with main dipole motion and westward drift of the non-dipole field respectively. These peaks can immediately be correlated with ours at 4000 and 2400–1850 yr. However, with the fuller, extended dataset containing three of the major swings, a predominant sense of looping can no longer be claimed. The retracing of the vector on the out and back parts of each swing (Fig. 6) contributes to both halves of the spectrum.

Although a non-dipole source for the 4000 yr variation is attractive, we cannot rule out the possibility that it originates in the main dipole. Indeed the phase relation between declination and inclination does not correspond with that for a westward drifting eccentric dipolar source. For such a source, pointing outward declination should move from a westerly to an easterly extremity as the source crosses the site longitude and produces extreme shallow inclinations, whereas our shallow inclinations coincide with peak easterly declinations. Spectral peaks close to our 2400 and 1850 yr values have been reported from many localities, e.g. Lake Michigan (Creer, Gross & Lineback 1976), Britain (Turner & Thompson 1981). It is almost certainly a feature of the non-dipole field, probably associated with the apparent westward drift.

The current literature contains much discussion of 'periodic' and 'non-periodic' secular variation. Datasets are generally short compared with the main features of the secular variation, containing only a few complete swings. Whether these swings appear regular or not is very critical on the density and accuracy of the dating control. For example, the originally suggested periodicity of the British post-glacial record (Thompson 1975) has been disproved by more intensive sampling and dating (Turner & Thompson 1981). A 'periodicity' corresponds to a regularly repeating phenomenon, running through the whole dataset, rather than a single occurrence of a major long-term feature occupying a significant fraction of the dataset, which may also produce a characteristic peak in the spectrum. Thus, when interpreting spectra of short palaeomagnetic datasets it is important to refer back to the original magnetograms to distinguish these two cases. In the present study the large amplitude of the 4000 yr peak corresponds to a cyclicity visible throughout the record.

6 Conclusions

The sequence of 212 stratigraphic horizons sampled spans some 12 000 yr starting about 31 000 bp and the sediments involved carry a stable remanence which has high within horizon precision. The following points emerge:

(1) Declination and inclination swings are of typical amplitudes (45° and 25° respectively) attributable to 'normal' secular variation at a latitude of 50°.

(2) There is no sign of any large-scale geomagnetic 'excursion' although it was anticipated that the so-called Mono Lake Excursion would probably be recorded at this site. Mono Lake is only 12° south, and the theoretical models of Liddicoat & Coe (1979) strongly imply that a large signal would be expected at our site. In conjunction with other published results this argues for the Mono Lake Excursion being placed prior to 36 000 bp.

(3) The secular variation pattern observed consists of relatively quiet intervals interspersed with major perturbations roughly every 4000 yr. During these perturbations the field vector swings systematically towards shallow inclinations and easterly declinations. Superimposed on this is a smaller amplitude variation with a time-scale of about 2000 yr.

(4) These perturbations lead to a bias such that the time-averaged field direction (D = 5.8°E, I = 64.2°, α_{95} = 0.9°, N = 212 horizons) is approximately 3° shallower and 6° more easterly than an axial dipole field (ADF, D = 0°, I = 67.5°), and thus yields a VGP which is both 'far-sided' and 'right-handed'. If the three low inclination swings are removed then the

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average remanence vector of the remaining horizons ($D = 1.2°$E, $I = 67.2°$, $\alpha_{95} = 0.8°$, $N = 125$ horizons) is indistinguishable from the ADF.

(5) The Bessette Creek dataset suggest that Cox's (1975) concept of spatially non-isotropic secular variation provides an appealing physical model for producing shallow inclinations and hence 'far-sided' poles.

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